MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 0.250 DIAMETER, SEMIRIGID, 50 OHMS, M17/129-RG401 AND M17/129-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

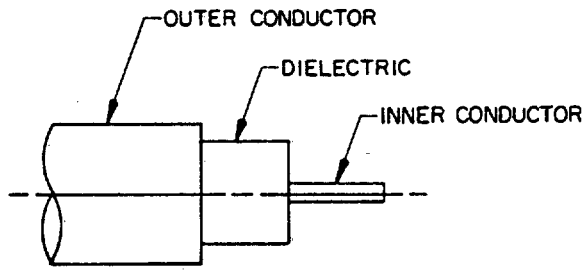


FIGURE 1. Configuration.

TABLE I. Description.

Part number	Inner conductor	Dielectric core	Outer conductor
M17/129-RG401	Solid, silver-coated copper wire Diameter, .0641 inch ±.0010	Type F-1 Diameter, .209 inch ±.002	Seamless copper tubing Diameter, .250 inch ±.001
M17/129-00001	Same as above	Same as above	Seamless copper tubing Diameter, .250 inch +.002, -.001 Tin plated per MIL-T-10727, Type I, .0003 to .0005 inch thick.

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Engineering information:

Continuous working voltage: 3,000 Vrms, maximum.
Operating frequency: 18 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -40° to +90°C.
Inner conductor properties:
DC resistance (maximum at 20°C): .257 ohms per 100 feet.
Elongation: 25 percent, minimum, per 10-inch specimen.
Engineering notes: This cable useful in critical RF performance applications (see connector series TNC per MIL-C-39012). This cable manufactured in 20-foot lengths.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Eccentricity: 7.5 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 4 pounds minimum; 100 pounds maximum.
Aging stability: Not applicable.
Stress crack resistance: Not applicable.
Outer conductor integrity: +200° ±5°C.
Cold bend: Not applicable.
Dimensional stability: +125° ±5°C.
Core from jacket: .015 inch, maximium.
Contamination: Not applicable.
Bendability: Mandrel diameter .75 inch, maximum.
Flammability: Not applicable.
Weight: 10.5 pounds per 100 feet maximum (M17/129-RG401).
10.6 pounds per 100 feet maximum (M17/129-00001).

Electrical:
Test frequency: 400 MHz to 18 GHz.
Spark test: Not applicable.
Voltage withstanding: 7,500 Vrms.
Insulation resistance: Not applicable.
Corona extinction voltage: 3,000 Vrms, minimum.
Characteristic impedance: 50.0 ohms ±0.5.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 29.6 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Phase stability: Not applicable.
Mechanically induced noise voltage: Not applicable
Time delay: Not applicable.

Part number: See table II.

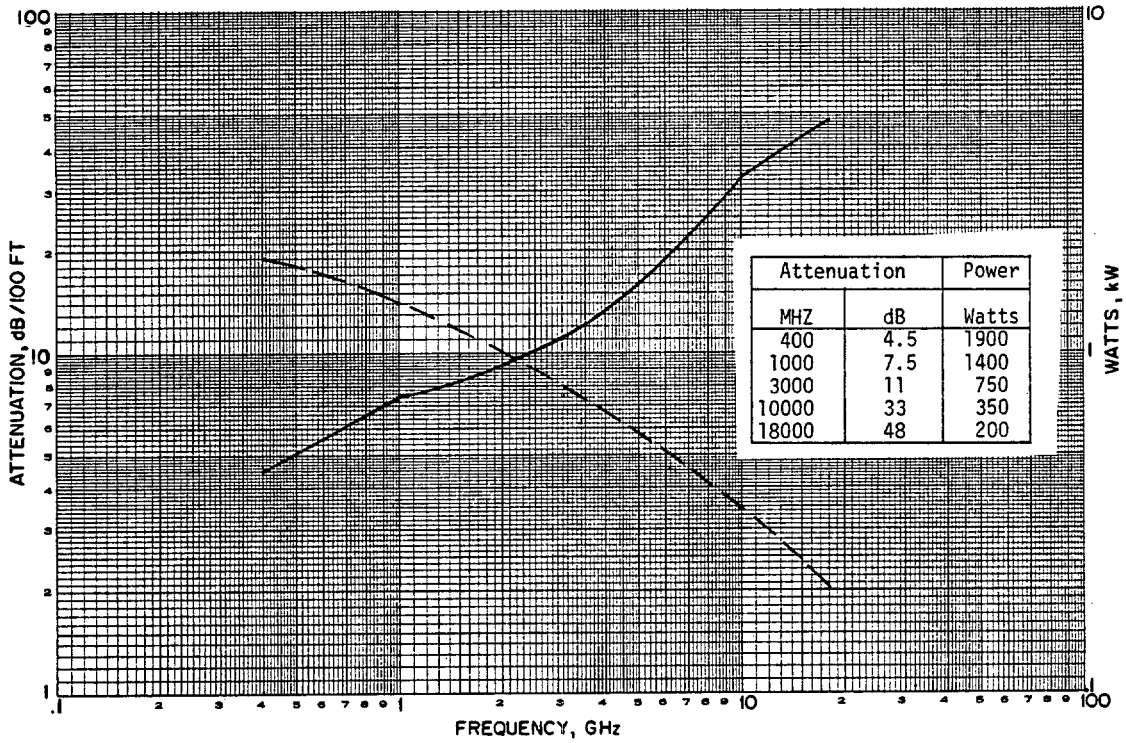
Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/129-RG401	RG-401/U
M17/129-00001	---

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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MAXIMUM ATTENUATION —— (Test requirements shall be as noted as line indicated on graph).
MAXIMUM POWER – – –
AT 25°C SEA LEVEL

FIGURE 2. Power rating and attenuation.

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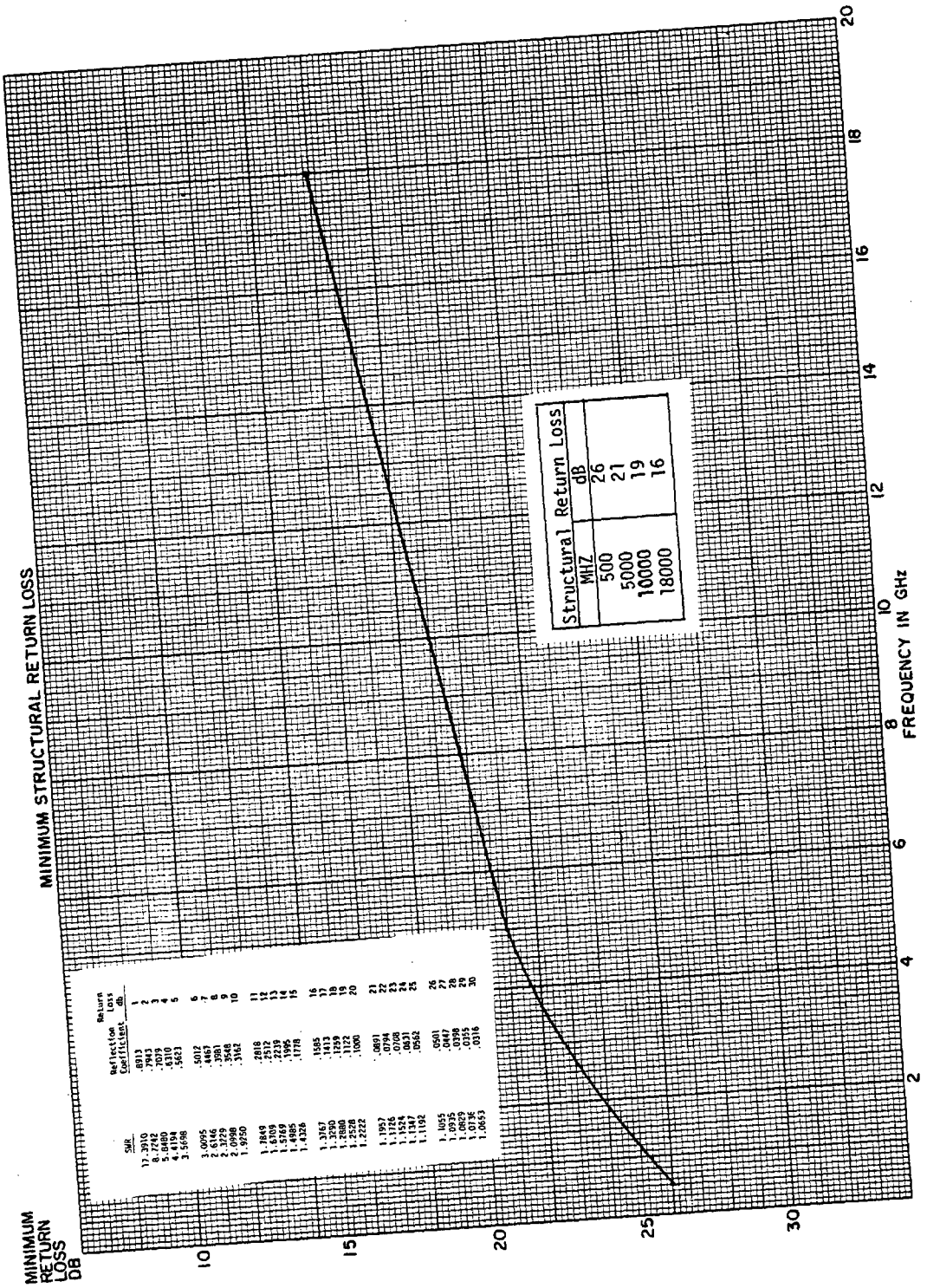


FIGURE 3. Structural return loss.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-0716-19)

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